

Lab 1 Solutions

Statistics for Laboratory Scientists

1.

```
> rep(1:5, 5:1)
[1] 1 1 1 1 1 2 2 2 2 3 3 3 4 4 5
```

 2.

```
> mat[!is.na(mat[,4]) & mat[,4] < 9,]
      method velo.rest velo.vo diam.rest diam.vo
2  capillaroscopy      NA    0.218      7.8    11.7
23 capillaroscopy    0.621    0.257      8.8     8.8
31   OPS imaging    0.557    0.110      8.7    10.6
42   OPS imaging    0.513    0.109      7.5     8.8
47   OPS imaging    0.386    0.049      8.1    11.2
52   OPS imaging    0.821    0.250      7.5     9.3
```

 3.

```
> sum(log10( seq(2, 1000, by=2) ))
[1] 1284.601
```

 4.

```
> f <- c(50, 65, 80, 95)
> cel <- 5*(f-32)/9
> cel
[1] 10.00000 18.33333 26.66667 35.00000
```

 5.

```
> sapply(mat[, -1], tapply, mat[, 1], mean, na.rm=TRUE)
      velo.rest  velo.vo diam.rest  diam.vo
OPS imaging    0.7365667 0.1523214 11.18621 12.13448
capillaroscopy 0.7732381 0.1961818 11.23704 12.03333

> sapply(mat[, -1], tapply, mat[, 1], sd, na.rm=TRUE)
      velo.rest  velo.vo diam.rest  diam.vo
OPS imaging    0.2766501 0.09187957 1.976419 1.817038
capillaroscopy 0.2392448 0.13684424 1.593773 1.633307
```
- (a) 0.20 (0.14)
 (b) 0.74 (0.28)
 (c) 11.2 (1.6)
 (d) 12.1 (1.8)